Database Design Document (DBD) Template

Reference Template for Document Validation System

Use this template as the standard reference structure for validating Database Design Document submissions. The validation system can compare uploaded DBD documents against these headings, required sections, order, and completeness.

Document Control

Document Title	Database Design Document
System Name	<Enter System Name>
Version	1.0
Prepared By	<Enter Name>
Date	<Enter Date>

1. Introduction

1.1 Purpose

Describe the purpose of this Database Design Document. Mention the system name and explain that this document defines the database structure, tables, relationships, constraints, and data storage requirements.

1.2 Scope

Describe the scope of the database design. Explain what parts of the system database are covered and mention any areas that are outside the scope.

1.3 Definitions, Acronyms, and Abbreviations

List important database-related terms, acronyms, and abbreviations used in this document.

1.4 References

List related documents, standards, diagrams, or resources used to prepare this database design document.

1.5 Document Overview

Briefly explain how the rest of this document is organized.

2. Database Overview

2.1 Database Purpose

Explain why the database is needed and what type of data it will store.

2.2 Database Type

Mention the type of database used, such as relational database, NoSQL database, or cloud database.

2.3 Database Management System

Mention the selected DBMS, such as MySQL, PostgreSQL, SQL Server, MongoDB, or Oracle.

2.4 High-Level Database Description

Provide a short overview of the main data areas/modules handled by the database.

2.5 Assumptions and Dependencies

List assumptions and dependencies related to the database design, such as required DBMS version, server availability, or application integration needs.

3. Data Requirements

3.1 Data Sources

Describe where the data comes from, such as user input, uploaded documents, admin records, APIs, or system-generated data.

3.2 Data Inputs

List the main input data that will be stored in the database.

3.3 Data Outputs

List the main output data retrieved or generated from the database.

3.4 Data Retention Requirements

Describe how long data should be stored and when it may be archived or deleted.

3.5 Data Volume Estimation

Estimate the expected amount of data, number of users, records, or transactions if available.

4. Conceptual Database Design

4.1 Main Entities

List the main entities in the system.

Entity Name	Description	Main Attributes
User	Stores user account details	user_id, name, email, role
Document	Stores uploaded document records	document_id, title, type, status
Validation Result	Stores document validation results	result_id, score, missing_sections, status

4.2 Entity Descriptions

Describe each entity and its purpose.

4.3 Entity Relationship Overview

Explain the main relationships between entities.

4.4 ER Diagram

Insert the Entity Relationship Diagram here.

5. Logical Database Design

5.1 Table List

List all database tables or collections used in the system.

Table Name	Purpose	Related Module
users	Stores user information	User Management
documents	Stores uploaded document information	Document Upload
validation_results	Stores validation outputs	Document Validation

5.2 Table Descriptions

Describe the purpose of each table or collection.

5.3 Attributes and Data Types

Define the fields, data types, sizes, and descriptions for each table.

5.4 Primary Keys

Identify the primary key of each table.

5.5 Foreign Keys

Identify foreign keys and the referenced tables.

5.6 Relationships

Describe one-to-one, one-to-many, and many-to-many relationships.

5.7 Normalization

Describe the normalization level applied, such as 1NF, 2NF, or 3NF, if applicable.

6. Physical Database Design

6.1 Database Schema

Provide the physical database schema or final table structure.

6.2 Table Structures

Provide detailed table structures including column name, data type, constraints, and description.

Column Name	Data Type	Size	Constraint	Description
id	INT	11	Primary Key	Unique record identifier
name	VARCHAR	100	NOT NULL	Name value
created_at	DATETIME	-	DEFAULT CURRENT_TIMESTAMP	Record creation date

6.3 Indexing Strategy

Describe indexes used to improve search and query performance.

6.4 Views

List any database views used in the system.

6.5 Stored Procedures and Functions

List stored procedures, database functions, or triggers if used.

6.6 Backup and Recovery Design

Describe backup methods, recovery process, and restore requirements.

7. Data Integrity and Constraints

7.1 Entity Integrity

Describe how primary keys and unique records are maintained.

7.2 Referential Integrity

Describe how foreign key relationships are maintained.

7.3 Domain Constraints

Mention valid data formats, allowed values, data ranges, and validation rules.

7.4 Business Rules

List business rules that affect the database design.

7.5 Error Handling for Invalid Data

Describe how invalid, duplicate, missing, or inconsistent data will be handled.

8. Security Requirements

8.1 User Access Control

Describe user roles and database access permissions.

8.2 Authentication and Authorization

Explain how access to database-related features is controlled.

8.3 Data Encryption

Mention encryption requirements for sensitive data at rest and in transit.

8.4 Audit Logging

Describe what database activities should be logged.

8.5 Privacy Requirements

Mention how personal or sensitive data should be protected.

9. Performance and Maintenance Requirements

9.1 Query Performance Requirements

Describe expected query response time or performance requirements.

9.2 Scalability Requirements

Describe how the database should handle growth in users, records, or transactions.

9.3 Maintenance Plan

Mention maintenance activities such as cleaning old records, optimizing tables, and monitoring performance.

9.4 Migration Requirements

Describe any requirements for importing, exporting, or migrating data.

9.5 Monitoring Requirements

Describe how database health, storage usage, and performance will be monitored.

10. Validation Rules

10.1 Required Sections

The uploaded DBD document should include the main required sections listed in this template.

- Introduction
- Database Overview
- Data Requirements
- Conceptual Database Design
- Logical Database Design
- Physical Database Design
- Data Integrity and Constraints
- Security Requirements
- Performance and Maintenance Requirements
- Validation Rules
- Appendices

10.2 Validation Criteria

The document validation system should check whether the document contains required database design sections, follows the correct order, includes database tables/entities, and belongs to the correct document type.

10.3 Scoring Criteria

The system may use score ranges such as 75-100 Accept, 50-74 Needs Improvement, and 0-49 Reject.

Score Range	Validation Result
75-100	Accept
50-74	Needs Improvement
0-49	Reject

11. Appendices

11.1 Appendix A: Sample ER Diagram

Insert a sample ER diagram or database relationship diagram.

11.2 Appendix B: Sample Table Structure

Insert sample table structures if available.

11.3 Appendix C: Additional Information

Include any extra database design information related to the system.